

AI Teacher – Technical Design Document (Version 1)

Project Vision

Build an AI-powered teacher that teaches from uploaded documents using a RAG pipeline, supports live voice interaction, allows student interruptions, and uses specialized AI agents for quizzes, examples, diagrams, summaries, resources, and progress tracking.

High-Level Workflow

- Student uploads PDFs/books.
- Documents are processed and indexed into the RAG pipeline.
- Main Teacher Agent retrieves content and starts teaching.
- Student may interrupt with voice at any time.
- ASR converts speech to text.
- Main Teacher Agent determines intent and routes to the correct sub-agent.
- Sub-agent returns its result.
- TTS speaks the response and teaching continues.

Agents

Main Teacher Agent

Coordinates teaching, retrieves RAG context, routes requests, maintains session.

Quiz Agent

Generates adaptive quizzes and evaluates answers.

Example Agent

Provides simpler explanations and real-world examples.

Diagram Agent

Creates diagrams and visual explanations.

Resource Agent

Suggests additional learning resources.

Summary Agent

Creates lesson summaries and revision notes.

Progress Agent

Tracks student progress and weak topics.

Recommended Tech Stack

- Frontend: Flutter
- Backend: Python + FastAPI
- Agent Framework: LangGraph

- LLM: Gemini
- ASR: AssemblyAI (prototype), Whisper (alternative)
- TTS: Gemini Live API (primary), ElevenLabs (optional)
- Vector Database: ChromaDB or Pinecone
- Database: PostgreSQL / Firebase Firestore
- Storage: Firebase Storage or Google Cloud Storage